Projects Portfolio

Ziming [Tiger] Ye **Mechatronics Engineering , University of Waterloo**

FRC 2019/2020 ROBOT

Unfortunately the final CAD is not available online.

Skills Demonstrated

Fusion 360

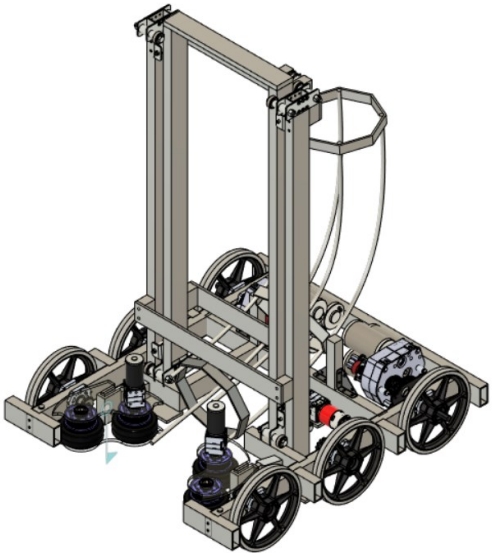
Machining: Drill Press, Filing, Milling, Band Saw

Communication

Leadership

Summary

Lead the 1246 FRC robotics team in the design and construction of the 2020 robot for the game Infinite Recharge. I was responsible for the overall design of the robot such as the elevator, power cell (balls) storage and final climb mechanism. Additionally I helped with the general integration of other components into the robot CAD and with allocating tasks for other team members for the construction of the robot.



CAD

The robot was designed in Fusion 360 around parts found in machine shop.

Fusion 360 simulation testing was done to ensure certain parts could handle loads required.

Integrated parts from McMaster-Carr into design of robot.

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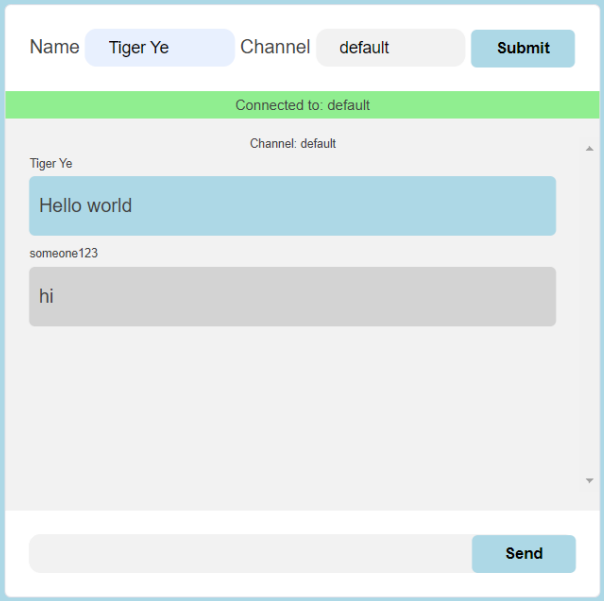
Chat App

Check it out by clicking the link above!

Or check out the code by pressing the link below.

https://github.com/tigerqye/chat_app



Skills Demonstrated

Javascript, CSS, HTML

Ably Realtime API

Node.js

Heroku

jQuery

Summary

Used Node.js to create a chat application which allows you to choose a name and channel which the user will then be connected to. This application allows you to send messages to anyone online using the Ably Realtime API while being hosted on Heroku.

Code

Used the Ably Realtime API through online tutorials and used it to send message objects to others that are subscribed to the same channel.

Deployed code to Heroku servers and set environment variables.

Used jQuery to edit html and add messages to the chat.

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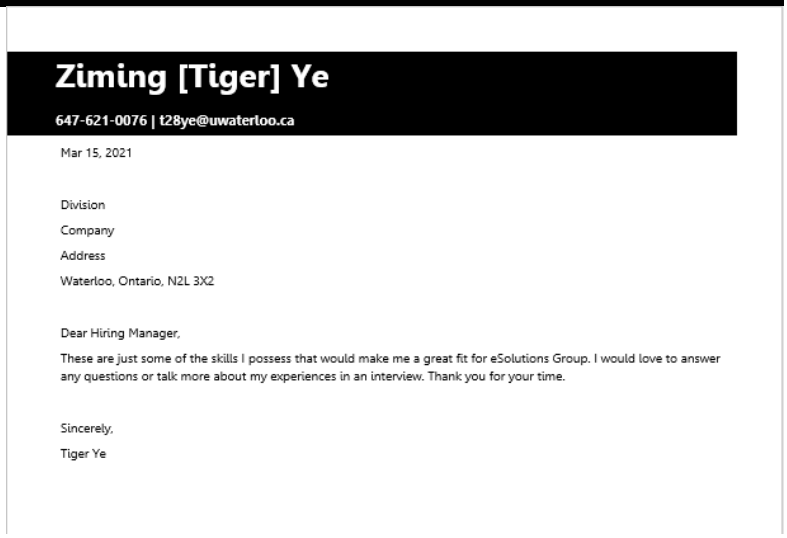
COVER LETTER SCRIPT

Check out the code by clicking the link above!

*being improved every job cycle :)



Ziming [Tiger] Ye

647-621-0076 | t28ye@uwaterloo.ca

Mar 15, 2021

Division
Company
Address
Waterloo, Ontario, N2L 3X2

Dear Hiring Manager,

These are just some of the skills I possess that would make me a great fit for eSolutions Group. I would love to answer any questions or talk more about my experiences in an interview. Thank you for your time.

Sincerely,
Tiger Ye

Skills Demonstrated

Python

Selenium

XPath

Chrome Driver

Python-docx

Summary

Setting up a cover letter can be tedious and time consuming leaving less time to actually express yourself through the letter. That is why I wrote a python script using Selenium that goes to WaterlooWorks and gets all the relevant information on a cover letter such as company name, postal code and more. It then uses Python-docx to write to a premade template Word document and saves it to a separate folder.

Code

Used Selenium and the chrome web driver to interact with Chrome webpages.

Utilized XPath to find elements on webpages and interact with them.

Searched through job posting to find key information such as company name, postal code and wrote to Word document.

Used Python-docx to write to Word files from Python.

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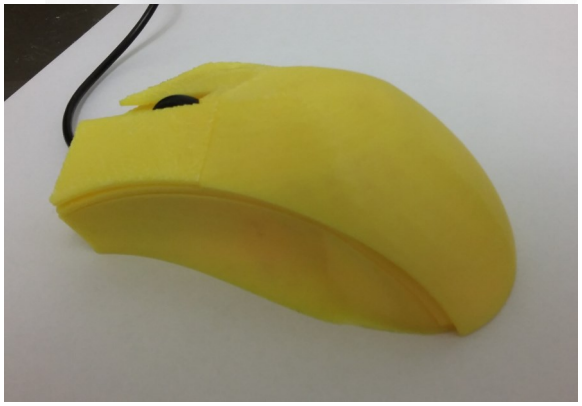
ROBOT 3D PRINTED COMPUTER MOUSE

Check out the CAD by clicking the link above!



Skills Demonstrated

Solidworks Surfacing Tools

Design for 3D Printing

Designing Under Constraints

Measuring Dimensions with Caliper

Cura

Summary

Used Solidworks to design a computer mouse body for the circuit board from another mouse. Used surfacing tools to design an elegant and ergonomic mouse body. Precise measurements were taken of the circuit board and the mouse was designed around those measurements.

CAD

Learned Solidworks surfacing tools such as 3D sketch, create plane, surface loft and boundary surface to aid in design of mouse.

Parts were designed with ease of 3D printing in mind as Ender 3 Pro supports are tough and hard to remove.

Designed a dummy circuit board CAD through caliper measurements of actual circuit board.

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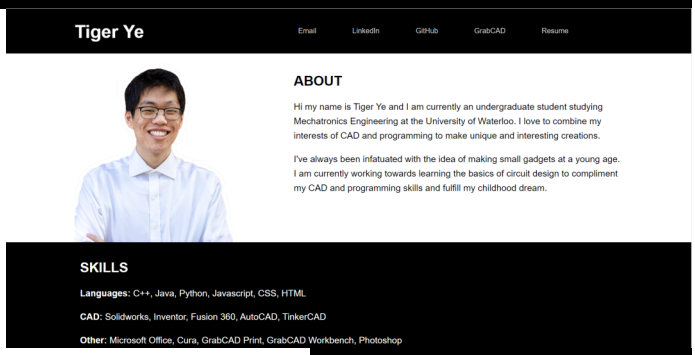
PERSONAL WEBSITE

Check it out by clicking the link above!

Or check out the code by pressing the link below.

https://github.com/tigerqye/tigerqye.github.io



Skills Demonstrated

JavaScript

CSS

HTML

Summary

Used JavaScript, CSS and HTML to build a responsive webpage to display personal projects. Responds to the screen and always ensures the best experience whether on computer or mobile. HTML, CSS and JavaScript were all learned through online resources such as YouTube videos and websites.

Code

Used JavaScript functions and querySelector to time the opening screen of website.

Used CSS flexbox for project boxes to ensure proper layout.

Utilized keyframes for subtle animations.

Used media screen to adjust images and text containers according to the width of the screen.

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CONVEYOR ASSEMBLY

Check out the CAD by clicking the link above!

Skills Demonstrated

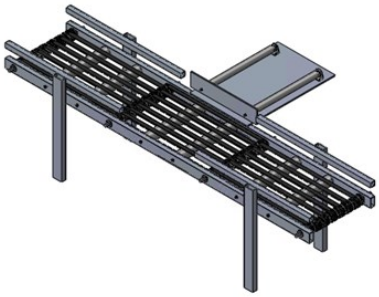
Solidworks

Part Integration

Summary

Designed a fully functional and feasible conveyor system that can sort boxes, scan for box orientation along with safety lights to ensure nobody gets to close to the conveyor while in operation.

CAD



Worked extensively with assemblies and sub-assemblies as was the nature of the modular design.

Learned to use the belt feature in Solidworks for the conveyor belt system.

Utilized mate tools to align parts to correct positions.

Analyzed torque-speed graph to find correct motor also taking into account continuous operating torque, duty cycle etc.

Designed around the goal of transporting 20x20x20cm and 10x10x30cm boxes with a max weight of 8.48kg.

Calculated desired rpm for given load.

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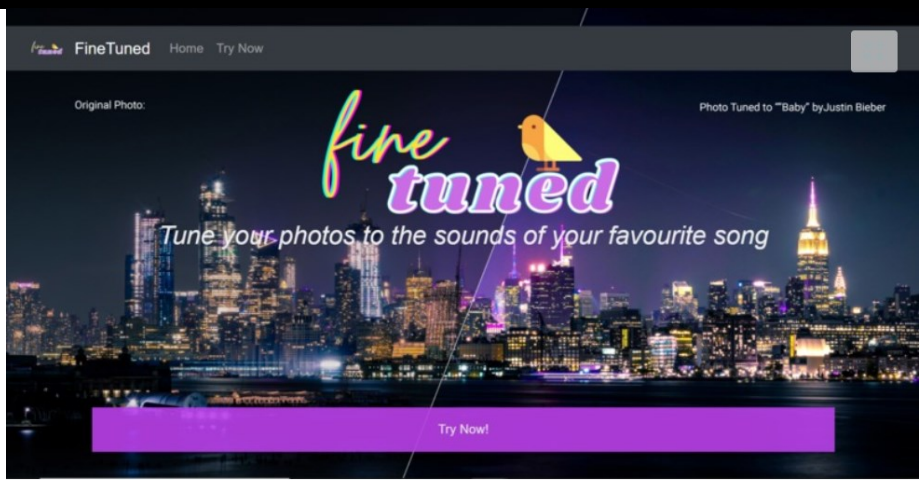
FineTuned

Check out the code by clicking the link above!



Skills Demonstrated

Python

PIL Library

Flask

Spotify API

JSON Formatting

Summary

Designed a webpage that edits user uploaded images based on song choice for Hack the North. Contrast, brightness and more will be adjusted to the tone of music inputted generating a unique aesthetic for your favourite songs.

Code

The application takes song information from the Spotify API and dynamically adjusts the image based on the song using Python.

Song information was given in JSON and specific parts were used for the editing algorithm.

Utilized the PIL Python library to edit the image.

The webpage is connected through a Flask backend.

Frontend was made using JavaScript, CSS and HTML.

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PROJECTS ROUNDUP

Curling Game

The first game I ever coded. Coded in Python using Pygame.

3D Printable Turntable

CADed a simple turntable for my mom to hold kitchen spices etc.

Game of Pig

Programed the game of pig for a culminating project. Coded in Java.

Phone Stand

Used Solidworks to design a phone stand which was later 3D printed.

IEEE754 Converter

Java application that manually converted integers to IEEE754 and reverse.

Millennium Puzzle

Used Fusion 360 to design a prop from a childhood tv show.

Linked List

Programed a linked list using Java.

Car CAD

Designed a simplistic car using surfacing features in Fusion 360.

Bank Class

Coded a bank class and bank account classes in Java.

For More Check Out My

GitHub: https://github.com/tigerqye

GrabCAD: https://grabcad.com/tiger.ye-1